



ELECTRONICS ENGINEERING • SEMICONDUCTOR DEVICES

BJT & FET

Characteristics, Biasing, Stability & Applications of Bipolar Junction and Field-Effect Transistors

BJT

Current-Controlled Device

JFET

Voltage-Controlled, Junction Gate

MOSFET

Voltage-Controlled, Insulated Gate



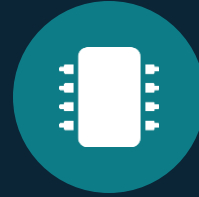
Agenda

PART 1 — BJT

- V-I Characteristics (Input & Output)
- DC Load Line and Operating Point
- Transistor as a Switch and an Amplifier
- Faithful Amplification
- Transistor Biasing and its Methods
- Transistor Stability and Stability Factor
- Testing and Transistor Datasheet

PART 2 — FET

- Types of FET: JFET & MOSFET
- Construction & Working of JFET
- Difference between BJT and JFET
- V-I Characteristics, Pinch-off Voltage
- Shorted-Gate Drain Current (I_{DSS})
- Parameters of JFET (g_m , r_d , μ)
- Biasing of JFET
- MOSFET Working & V-I Characteristics
- Difference between JFET and MOSFET



PART ONE

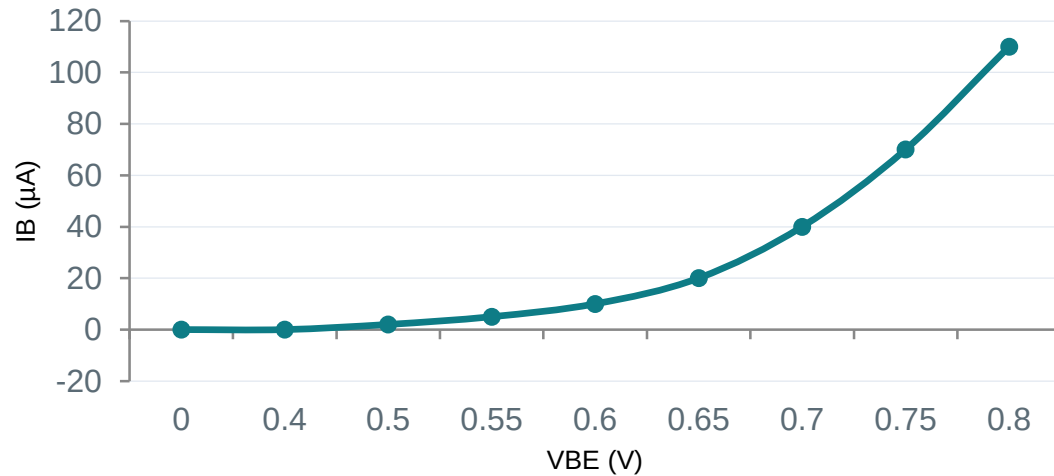
Bipolar Junction Transistor

Structure, characteristics, biasing and stability of the BJT

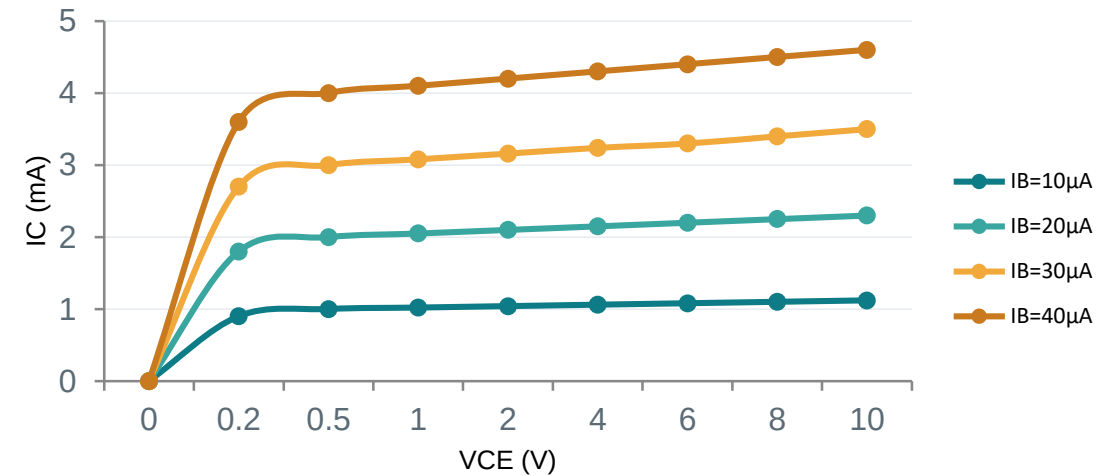


V-I Characteristics

Input Characteristic (CE mode)



Output Characteristic (CE mode)

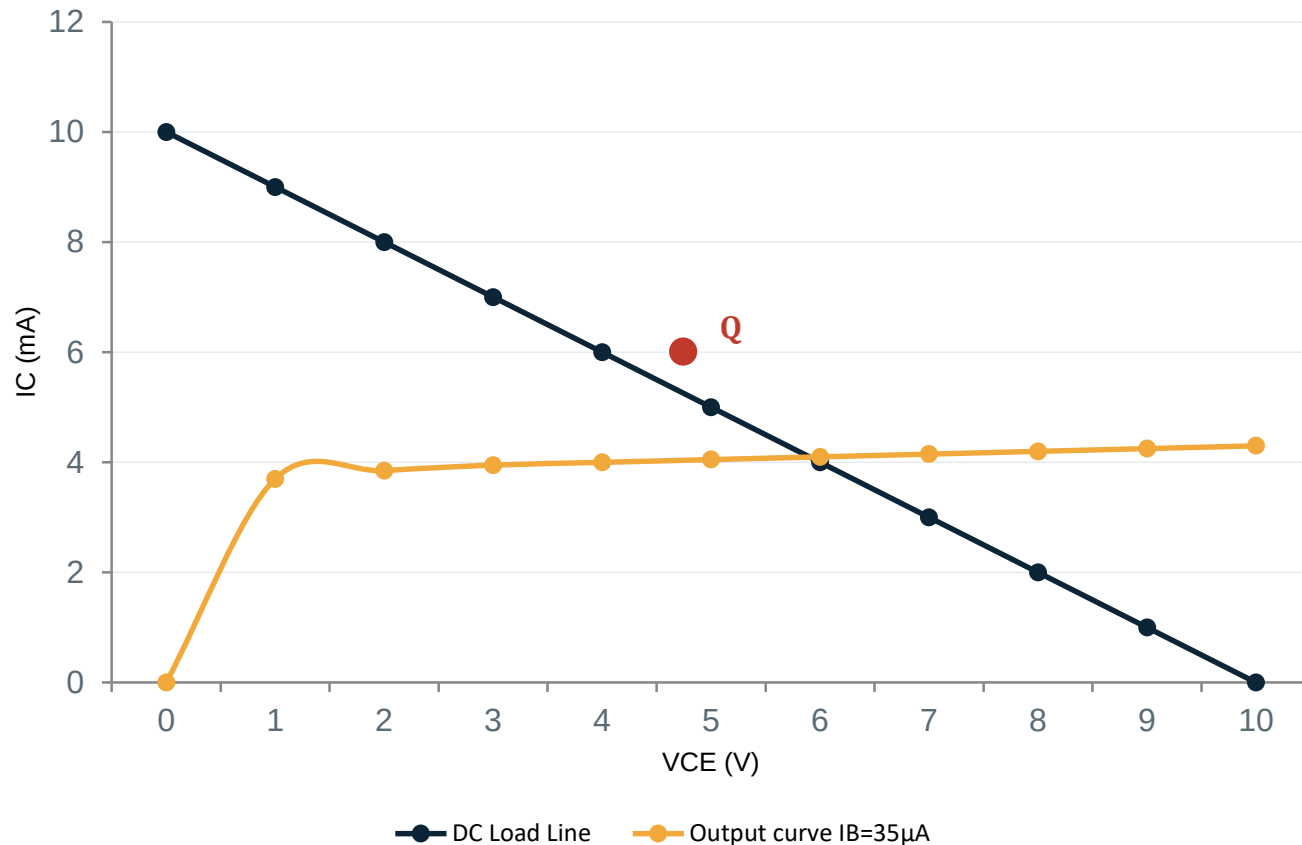


Three Regions of Operation

- Cut-off region: both junctions reverse biased — $I_C \approx 0$, transistor behaves as an OFF switch
- Active region: EB junction forward biased, CB junction reverse biased — $I_C \approx \beta \cdot I_B$, used for amplification
- Saturation region: both junctions forward biased — I_C is maximum and nearly independent of I_B , transistor behaves as an ON switch



DC Load Line and Operating Point



Load Line Equation

$$V_{CC} = I_C R_C + V_{CE}$$

A straight line on the output characteristics fixed by the supply voltage V_{CC} and load resistance R_C .

- Cut-off point: $V_{CE} = V_{CC}$, $I_C = 0$
- Saturation point: $I_C = V_{CC} / R_C$, $V_{CE} = 0$

Operating (Quiescent) Point

- Intersection of the DC load line with the output characteristic curve for a given I_B
- Represents zero-signal values of I_C and V_{CE}
- Must be at the centre of the load line for maximum, distortion-free (faithful) output swing



Transistor as a Switch

- Operates only in cut-off and saturation regions — the active region is skipped almost instantly
- A small base current drives the transistor between its two extreme states
- Forms the basis of digital logic gates, relay drivers and power-control circuits



OFF STATE — Cut-off

- $I_B = 0 \rightarrow$ both junctions reverse biased
- $I_C \approx 0$ (only leakage current)
- $V_{CE} \approx V_{CC}$ — acts as an OPEN switch



ON STATE — Saturation

- I_B is large enough to drive I_C to its maximum
- Both junctions forward biased
- $V_{CE} \approx V_{CE(sat)} \approx 0.2 \text{ V}$ — acts as a CLOSED switch



Transistor as an Amplifier

- Transistor is biased to operate in the active region, where it can raise the strength of a weak signal
- A small change in base current (ΔI_B) produces a much larger change in collector current (ΔI_C)
- Current amplification factor: $\beta = \Delta I_C / \Delta I_B$ (common-emitter)
- Voltage gain results because the output signal is developed across a high collector load resistance R_C
- $A_V = \beta \times (R_C / R_{in})$ — output voltage is 180° out of phase with the input in CE configuration

Why Amplification Works

- Input circuit (base-emitter) has low resistance — small voltage swing needed
- Output circuit (collector-emitter) has high resistance — same current now develops a large voltage
- Power gain = Current gain $\times$ Voltage gain

Key Amplifier Gain Relations

Gain	Expression	Meaning
Current gain (β)	I_C / I_B	Ratio of output to input current
Voltage gain (A_V)	$\beta \times R_C / R_{in}$	Ratio of output to input voltage
Power gain (A_P)	$A_V \times \beta$	Ratio of output to input power



Faithful Amplification

Faithful amplification is the process of raising the strength of a weak signal so that the output is an exact replica of the input, without any distortion.



Proper Zero-Signal Collector Current

A minimum collector current must flow even with no input signal so the transistor never enters cut-off during the negative half-cycle.



Proper Base-Emitter Bias

The emitter-base junction must remain forward biased ($\sim 0.5\text{ V}$ for Si) throughout the input cycle.



Proper Collector-Emitter Voltage

VCE must stay above $V_{CE(sat)}$ at all times so the transistor never enters saturation on the positive half-cycle.

Result of violating these conditions: clipping of the positive or negative half-cycle → distorted, unfaithful output. Correct DC biasing keeps the operating point centred on the load line, avoiding both cut-off and saturation.



Transistor Biasing

- Biasing: applying suitable DC voltages/currents to establish a fixed operating (Q) point in the active region
- Needed so the transistor amplifies the AC signal faithfully without clipping
- A good biasing circuit must:
 - Establish the Q-point at the centre of the DC load line
 - Keep the Q-point stable against temperature and transistor (β) variation
 - Be simple, economical and easy to design

Effect of Temperature on Q-Point

- Leakage current I_{CO} increases with temperature, roughly doubling every 10°C rise
- β (current gain) also increases with temperature
- V_{BE} decreases about 2.5 mV per $^{\circ}\text{C}$ rise
- Together these shift I_C upward — the Q-point can drift into saturation if left uncompensated
- A stable biasing network counteracts this drift automatically



Methods of Transistor Biasing

Fixed-Bias (Base Bias)

A single resistor R_B from V_{CC} to the base sets a constant I_B .

✓ Simple, few components

✗ Highly unstable — I_C depends directly on β

Collector-to-Base Bias

R_B is returned from the collector (not V_{CC}) to the base, providing negative feedback.

✓ Better stability than fixed bias

✗ Reduces voltage gain slightly; moderate stability

Voltage-Divider (Self) Bias

R_1 - R_2 divider fixes base voltage; emitter resistor R_E provides negative feedback.

✓ Most stable — nearly independent of β

✗ Needs more components

Emitter Bias (Dual Supply)

Uses $+V_{CC}$ and $-V_{EE}$ with R_E providing strong feedback.

✓ Excellent stability, used in IC/discrete linear design

✗ Requires a dual power supply



Transistor Stability and Stability Factor

Stability Factor (S)

$$S = \Delta I_C / \Delta I_{CO} \text{ (at constant } I_B \text{ and } \beta)$$

Defines the rate of change of collector current with respect to the reverse-saturation (leakage) current I_{CO} .

A lower value of S indicates better thermal stability of the Q-point.

Why Stability Matters

- I_{CO} , β and V_{BE} all vary with temperature and from device to device
- Without stabilisation the Q-point can drift into saturation or cut-off, causing distortion
- A well-designed bias network keeps I_C and V_{CE} nearly constant despite these variations

Typical Stability Factor by Biasing Method

Biasing Method	Stability Factor (S)	Remark
Fixed bias	$S = (1 + \beta)$ — highest	Poor stability
Collector-to-base bias	$S < (1 + \beta)$	Moderate stability
Voltage-divider bias	$S \approx 1$ (minimum)	Best stability
Emitter bias (dual supply)	S close to 1	Very good stability

Lower $S \Rightarrow I_C$ changes less for a given change in $I_{CO} \Rightarrow$ more stable operating point over temperature and device variation.



Testing and Transistor Datasheet

Testing a BJT (Multimeter Diode Test)

- A BJT is essentially two back-to-back PN junctions (base-emitter and base-collector)
- Set the multimeter to diode-test mode
- Base-Emitter junction: forward drop $\approx 0.6\text{-}0.7\text{ V}$ (Si); reverse direction should show open circuit (OL)
- Base-Collector junction: forward drop $\approx 0.6\text{-}0.7\text{ V}$ (Si); reverse direction should show OL
- Collector-Emitter path: should read open circuit in both directions (no direct junction)
- OL in both directions on any pair, or a low reading in both directions, indicates an open or shorted (faulty) transistor

Key Datasheet Parameters

Parameter	Meaning
$V_{CEO}(\text{max})$	Max collector-emitter voltage (base open)
$V_{CBO}(\text{max})$	Max collector-base voltage (emitter open)
$I_C(\text{max})$	Maximum continuous collector current
$P_D(\text{max})$	Maximum power dissipation
$h_{FE}(\beta)$	DC current gain, I_C / I_B
f_T	Transition (gain-bandwidth) frequency
$V_{CE}(\text{sat})$	Collector-emitter voltage at saturation



PART TWO

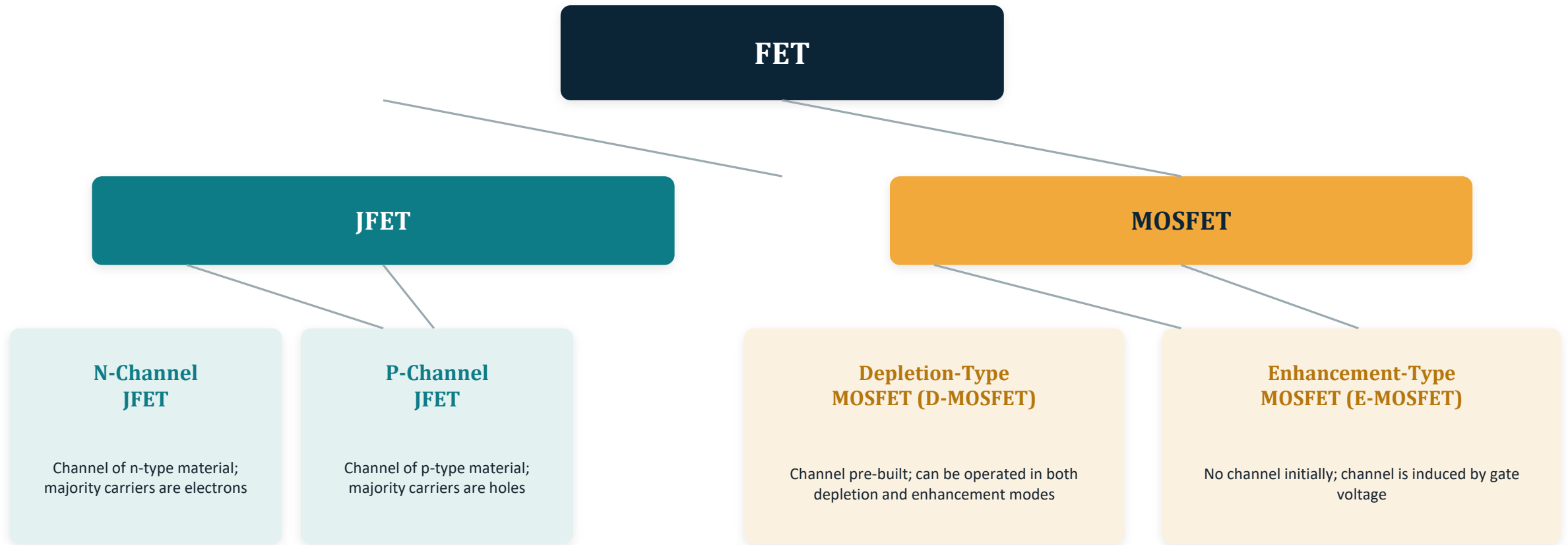
Field-Effect Transistor

Types, construction, characteristics and biasing of JFET and MOSFET



Types of FET

A Field-Effect Transistor is a unipolar, voltage-controlled device — current flows through a single type of charge carrier and is controlled by an electric field.





Construction and Working Principle of JFET

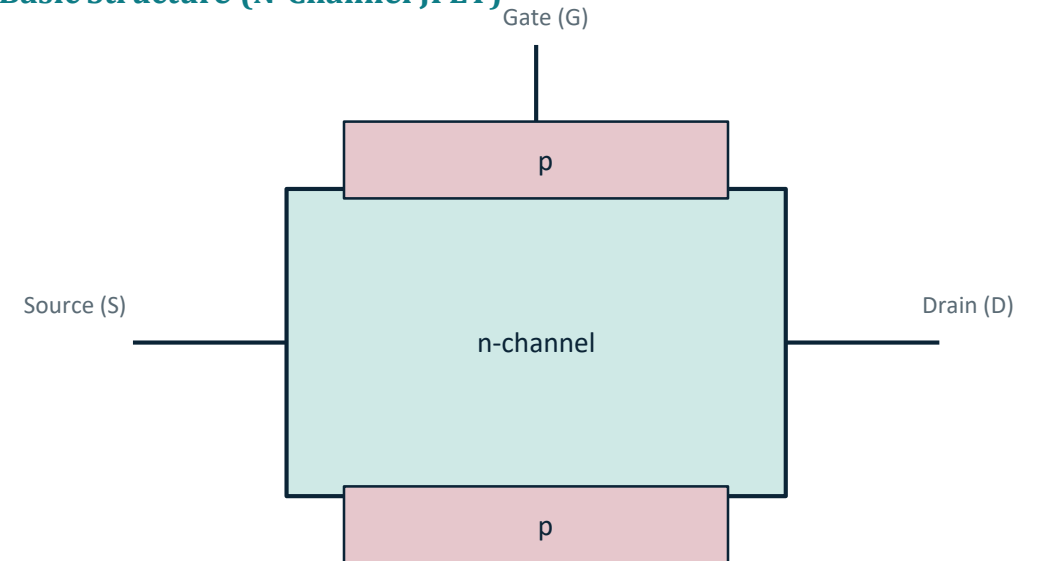
Construction

- A bar of lightly doped semiconductor (e.g. n-type) forms the channel between two ohmic terminals: Source (S) and Drain (D)
- Two heavily doped regions of the opposite type (p-type) are diffused on either side of the bar and joined together to form the Gate (G)
- The two gate regions are internally connected and behave as a single PN junction with the channel

Working Principle

- Gate-Source junction is always reverse biased — gate draws virtually no current (very high input impedance)
- Increasing reverse VGS widens the depletion region into the channel, narrowing it and reducing drain current I_D
- VDS moves electrons from source to drain through the narrowed channel

Basic Structure (N-Channel JFET)



The reverse-biased gate junctions create depletion regions (shown widening with VGS) that pinch the conducting channel.



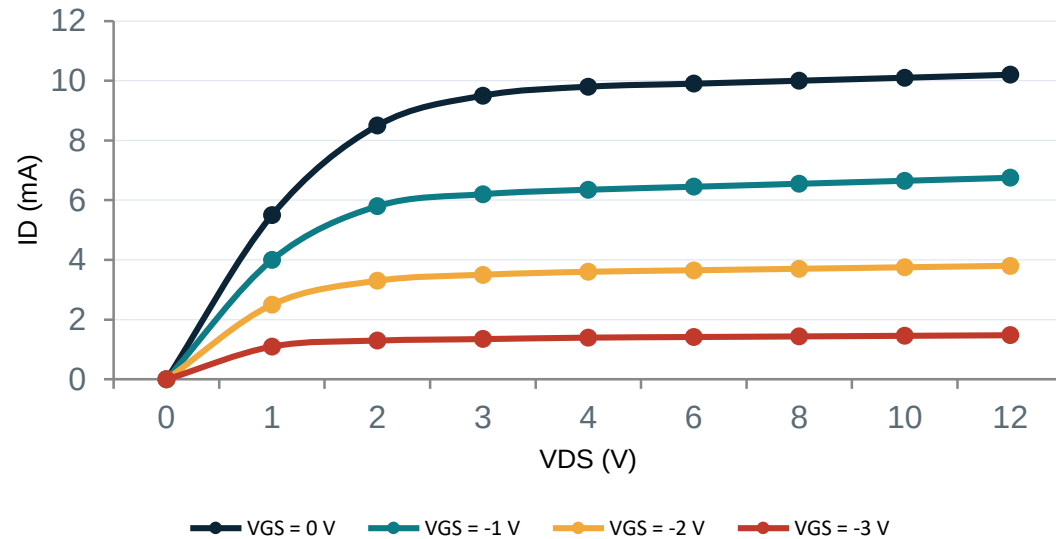
Difference between BJT and JFET

Parameter	BJT	JFET
Type of device	Bipolar — both electrons and holes conduct	Unipolar — only one type of carrier conducts
Control mechanism	Current-controlled (I_B controls I_C)	Voltage-controlled (V_{GS} controls I_D)
Input impedance	Low ($k\Omega$ range)	Very high ($M\Omega$ to $G\Omega$ range)
Terminals	Emitter, Base, Collector	Source, Gate, Drain
Gain parameter	Current gain, β	Transconductance, g_m
Noise	Higher, due to minority-carrier injection	Lower — majority carrier device
Thermal stability	Comparatively less stable	Better thermal stability
Switching speed	Moderate	Generally faster

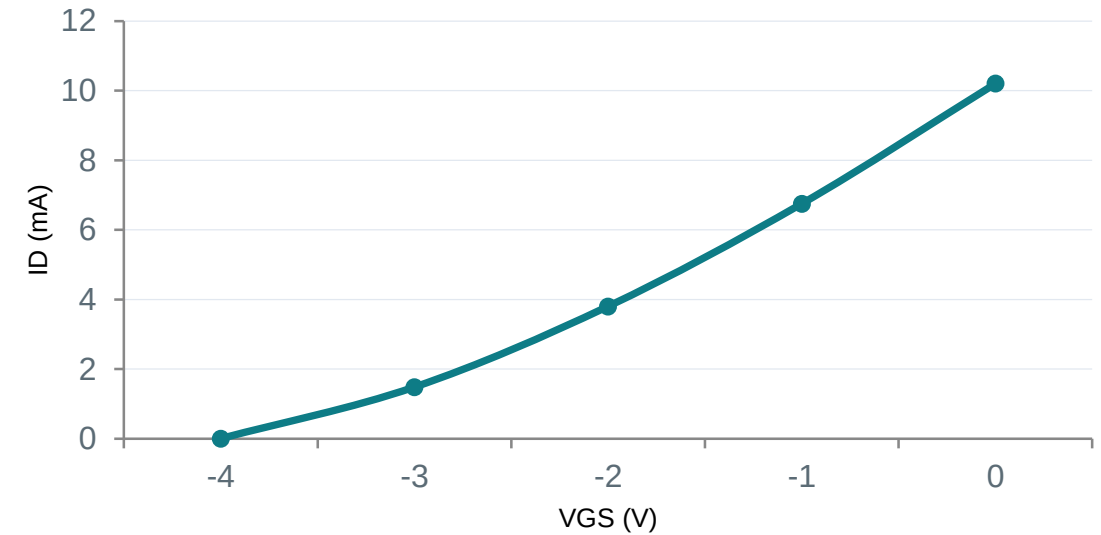


V-I Characteristics of JFET

Drain Characteristics (ID vs VDS)



Transfer Characteristics (ID vs VGS)



- Drain characteristics show three regions: ohmic (linear), saturation/pinch-off (nearly constant ID), and breakdown at high VDS
- Transfer characteristic follows Shockley's equation: $ID = IDSS (1 - VGS/VP)^2$, relating gate voltage to drain current



Pinch-off Voltage

Definition

- For a given V_{DS} , the pinch-off voltage V_P is the value of V_{DS} at which the depletion regions from both sides of the gate just meet at the centre of the channel
- Beyond this point, I_D becomes almost constant with further increase in V_{DS} — this marks the start of the saturation (constant-current) region
- It is also used to denote the gate-source voltage $V_{GS(off)}$ at which the channel is fully pinched off and I_D drops to (nearly) zero, for $V_{GS} = 0$ drain characteristic taken as reference

Key Points

- $|V_P| = |V_{GS(off)}|$ for the same JFET (numerically equal, opposite context)
- Operating the JFET beyond pinch-off (in the saturation region) is essential for its use as a constant-current source and as a linear amplifier
- V_P is a datasheet parameter, typically a few volts (e.g. -2 V to -8 V for common n-channel JFETs)
- Below pinch-off (small V_{DS}), the JFET behaves like a voltage-controlled resistor



Shorted-Gate Drain Current (IDSS)

Definition

- IDSS is the drain current that flows when the gate is shorted to the source ($V_{GS} = 0\text{ V}$) and V_{DS} is large enough to be in the saturation/pinch-off region
- It represents the maximum possible drain current for a given JFET, since any negative V_{GS} only reduces I_D further
- IDSS is a key datasheet rating used as the reference value in Shockley's equation

Shockley's Equation

$$I_D = I_{DSS} \left(1 - V_{GS} / V_P \right)^2$$

- I_D = drain current at a chosen V_{GS}
- I_{DSS} = drain current at $V_{GS} = 0$ (shorted gate)
- V_P (or $V_{GS(off)}$) = pinch-off voltage
- As V_{GS} becomes more negative (n-channel), I_D decreases from I_{DSS} toward zero at $V_{GS} = V_P$



Parameters of JFET

Transconductance (g_m)

$$g_m = \Delta I_D / \Delta V_{GS} \text{ (at constant } V_{DS})$$

Measures the control that gate voltage has over drain current; unit: siemens (S) or mA/V. Also called forward transfer conductance, g_{fs} .

AC Drain (Output) Resistance (r_d)

$$r_d = \Delta V_{DS} / \Delta I_D \text{ (at constant } V_{GS})$$

Resistance offered by the drain-source channel to AC signal; very high in the saturation/pinch-off region.

Amplification Factor (μ)

$$\mu = g_m \times r_d$$

Relates the maximum voltage gain obtainable from the JFET; equals the ratio $\Delta V_{DS}/\Delta V_{GS}$ at constant I_D .



Biasing of JFET

Fixed Bias

A separate negative supply V_{GG} applies a fixed reverse voltage directly to the gate through a high-value resistor R_G .

Simple; but needs an extra negative supply and gives poor Q-point stability

Self Bias

Source resistor R_S develops the gate-source voltage using the drain current itself; gate is returned to ground through R_G (no gate current, no drop across R_G).

Single supply; most common method for small-signal JFET amplifiers

Voltage-Divider Bias

A resistive divider (R_1 - R_2) sets the gate voltage; source resistor R_S provides additional feedback, similar in spirit to BJT voltage-divider bias.

Improved stability against I_{DSS} and V_P variation between devices

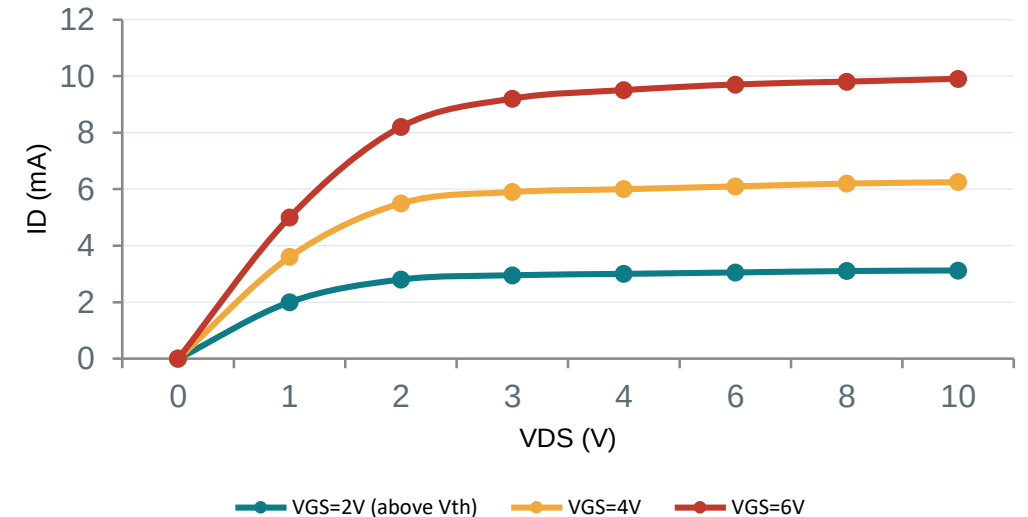


Working of MOSFET and V-I Characteristics

Construction & Working

- Gate is a metal (or polysilicon) electrode fully insulated from the channel by a thin layer of silicon dioxide (SiO_2) — no PN junction at the gate
- Extremely high input impedance since the gate draws essentially zero DC current
- Depletion-type MOSFET: channel is physically present at $V_{GS} = 0$; a negative V_{GS} depletes carriers (reduces I_D), a positive V_{GS} enhances the channel (increases I_D) — works in both modes
- Enhancement-type MOSFET: no channel exists at $V_{GS} = 0$ ($I_D \approx 0$); applying a V_{GS} beyond the threshold voltage $V_{GS(th)}$ induces (creates) a conducting channel between source and drain

V-I Characteristics



Shown: Enhancement-mode NMOS drain characteristics — I_D increases with V_{GS} above threshold, then saturates with V_{DS} .



Difference between JFET and MOSFET

Parameter	JFET	MOSFET
Gate structure	PN junction (reverse biased)	Metal gate insulated by SiO ₂ layer
Gate current	Very small reverse leakage current	Essentially zero (insulated gate)
Input impedance	High (MΩ range)	Extremely high (up to 10 ¹⁴ Ω)
Modes of operation	Depletion mode only	Depletion and/or Enhancement mode
Channel at V _{GS} = 0	Always exists	Exists in D-MOSFET; absent in E-MOSFET
V _{GS} polarity range	Reverse bias only (one polarity)	Can be operated with either polarity
Handling	Relatively robust	Static-sensitive — gate oxide can be damaged by ESD
Typical use	Low-noise amplifiers, analog switches	Digital ICs (CMOS), power switching



RECAP

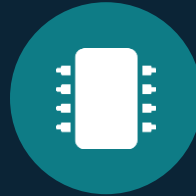
Summary

BJT — Key Takeaways

- Current-controlled, bipolar device with cut-off, active and saturation regions
- DC load line and Q-point set the operating conditions for switching or amplifying
- Faithful amplification demands correct, stable biasing
- Voltage-divider bias gives the lowest (best) stability factor

FET — Key Takeaways

- Voltage-controlled, unipolar device with very high input impedance
- JFET: PN-junction gate, depletion-mode only; MOSFET: insulated gate, depletion and/or enhancement mode
- Key parameters: pinch-off voltage V_P , shorted-gate current I_{DSS} , transconductance g_m
- MOSFETs dominate modern digital and power electronics (CMOS)



Thank You

Questions & Discussion — BJT & FET: Characteristics, Biasing and Applications